

Invasive plants detection and distribution patterns analysis through self-attention enhanced semantic segmentation in UAV imagery and Moran's index

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ABSTRACT

The development of sustainable agriculture necessitates the rapid identification and efficient removal of invasive plants. Traditionally, the investigation of invasive plants relies on manual surveys, which are prone to subjective errors and require significant human labor. This study introduces an innovative approach for accurately identifying and efficiently eliminating invasive plants. High-resolution aerial images were captured using drones, and a novel semantic segmentation network, based on DeepLabV3+ and the Self-Attention mechanism, was developed to reliably identify two globally distributed invasive species: *Erigeron annuus* (L.) Pers. and *Erigeron canadensis* L. at the pixel level. Experimental results from orchard imagery revealed that the proposed approach demonstrated remarkable performance, achieving a mean Precision (mPrecision) of 0.972, a mean Intersection over Union (mIoU) of 0.947, and a mean Pixel Accuracy (mPA) of 0.973 for its overall effectiveness. A high-resolution species distribution map at pixel-level was generated using the results of this model. The study further explored ecological analysis methods of this distribution map, and successfully calculated the coverage area, coverage rate, global Moran's index, and local Moran's index of the two invasive plants. The findings revealed that the intraspecific distribution patterns of both species are characterized by clustering, with Global Moran's Index values of 0.03 for *Erigeron annuus* (L.) Pers. and 0.23 for *Erigeron canadensis* L. The clustering map facilitates the rapid identification of invasive plant cluster centers, enabling more targeted weed control measures. The efficient pixel-level invasive plant identification model and species distribution pattern analysis proposed in this study holds significant implications for agricultural production and ecological surveys. They support precise and rapid invasive plant control and to reduce pesticide use. The proposed method can also be implemented on other platforms to provide fast, flexible, and accurate invasive plant mapping and precision agriculture applications.

1. Introduction

Invasive species are alien species introduced into non-native environments. They grow and reproduce rapidly, often negatively impacting the local ecosystem, economy, and human life (Peller and Altermatt, 2024). The expansion of globalization and increasing human activities have exacerbated the spread of invasive plants (Weidlich et al., 2020). In agricultural production, human farming operations provide favorable

conditions for the spread of invasive alien plants (Tataridas et al., 2022). Water hyacinth (*Pontederia crassipes*), originally introduced as an ornamental plant in several countries, has become a pervasive invasive species in rice fields (Bazie et al., 2022). The extensive use of fertilizers in paddy fields provides an abundant nutrient supply, creating an ideal environment for its proliferation. Additionally, agricultural irrigation systems serve as efficient dispersal pathways, enabling the plant to establish in irrigation channels and subsequently spread to natural

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waterways. Similarly, in soybean fields, large-scale monoculture practices weaken the resilience of local ecosystems, facilitating the widespread invasion of redroot pigweed (*Amaranthus retroflexus* L.). The frequent transportation of soybeans and their derivatives further accelerates the dissemination of this weed to new agricultural regions (Gawęda et al., 2020). Due to invasive alien plants' strong ecological adaptability, reproductive capability, and allelopathic effects, these alien plants directly affect the yield and quality of crops, leading to significant disruptions in agricultural production. For instance, ragweed (*Ambrosia artemisiifolia*) grows rapidly and produces a large number of seeds. Its pollen can cause allergic reactions and asthma in humans (Montagnani et al., 2023). Ragweed, with its strong reproductive capacity and competitive advantage, poses a substantial threat to crop yields. In sunflower fields, a density of ten ragweed plants per square meter can result in a 37 % yield reduction (Knolmajer et al., 2024). In soybean fields, even a single ragweed plant per square meter can decrease yield by 18 %. When five ragweed plants are present in a pot experiment, soybean yield losses can reach an average of 84 % compared to the weed-free control (Hall et al., 2021). Moreover, it has developed resistance to various herbicides, necessitating specific weed control methods for its eradication (Loubet et al., 2021). Similarly, Japanese knotweed (*Reynoutria japonica*) is highly invasive. It reproduces through rhizomes and its robust root system disrupts soil structure (Stefanowicz et al., 2021). Research indicates that it releases phenolic compounds that alter soil properties and inhibit the growth of other plants, which leads to ecosystem degradation and reduced biodiversity (Stefanowicz et al., 2022). This further decreases agricultural yields. Additionally, it can penetrate and damage agricultural infrastructure, hindering farming operations. Its strong ecological adaptability and resistance to many herbicides make it difficult to eliminate (Martin et al., 2020). Both *Erigeron annuus* (L.) Pers. and *Erigeron canadensis* L. are globally distributed invasive plants originating from North America, belonging to the Asteraceae family. They are now widely found in Asia, Europe, Oceania, and Africa, and have been listed as invasive species in multiple countries (Song et al., 2018; Yan et al., 2020). Additionally, studies have shown that extracts from both plants exhibit allelopathic effects, interfering with the growth of other plant species (Kudryavtseva et al., 2020). Their strong ecological adaptability, resistance to herbicides, and allelopathic effects significantly impact agricultural production. Therefore, accurately monitoring the distribution of these plants and analyzing their spatial distribution patterns are essential for the development of effective and targeted weed control strategies.

The primary method for understanding the distribution of invasive plant species is manual surveying. This process involves establishing multiple sample plots. For species identification, surveyors depend on the key morphological characteristics of plants within these plots, supplemented by botanical knowledge, reference books, and herbarium specimens. For species that are difficult to identify on-site, samples need to be collected and taken back to the laboratory for expert identification. In the investigation of vegetation cover, direct observation methods are typically used to visually estimate the area proportion each plant occupies within the plot. Alternatively, a grid method can be used, where a grid is set up within the plot, and the plant species touching the grid points are recorded to estimate coverage (Bhat et al., 2020). Regardless of the method used, the accuracy of plant identification and coverage estimation heavily depends on the surveyor's experience. This process inevitably leads to errors while also requiring a significant amount of labor and time (Lisner and Lepš, 2020; Morrison, 2021). Satellite remote sensing images have been proven successful for larger-scale research (Royimani et al., 2019). However, due to their spatial and temporal resolution constraints, challenges arise in accurately determining community species composition, precise positioning, and efficient weed management (Xue and Su, 2017). Unmanned Aerial Vehicles (UAVs) have become an effective platform for obtaining higher spatial resolution images. They provide flexibility in controlling flight altitude and trajectory to meet specific data collection requirements, and support a

diverse range of camera payloads, including RGB, multispectral, hyperspectral, thermal infrared, 3D LiDAR, and other types digital sensors. RGB sensors are cost-effective, easy to use, widely available, and produce results that are easy to interpret. Combined with the flexible flight capabilities of UAV platforms, RGB sensors can maximize their advantages, making them widely used in vegetation species identification, monitoring plant growth, detecting diseases, and identifying pests (Dutta and Goswami, 2020). Furthermore, there's potential for the further development of drones for precision monitoring and spraying (Esposito et al., 2021).

From traditional machine learning to deep learning-based network models, significant achievements have been made in vegetation detection using high-resolution aerial imagery (Kattenborn et al., 2021; Turhal, 2022; Zhao et al., 2023). Deep learning offers high accuracy, automatic feature extraction, and the ability to process large-scale data in vegetation recognition (Ferchichi et al., 2022). It provides robust identification even in complex environments (Haq et al., 2023). Given sufficient data, deep learning methods can generate and infer new features without explicit instructions on which features to use or how to extract them, enabling the construction and utilization of more abstract concepts (Behera et al., 2023; Retallack et al., 2022). Compared to manual field surveys, deep learning not only reduces the difficulty of weed identification, but also provides high accuracy and stability (Parra et al., 2020). With the further development of image segmentation models, an increasing number of studies are utilizing pixel-level segmentation methods (Deng et al., 2020; Kamath et al., 2022). Unlike traditional CNNs that classify entire images, these segmentation networks can precisely delineate the shape and contours of weeds at the pixel level, accurately depicting their distribution rather than just indicating the general location of the plants (Gupta et al., 2023; Wang et al., 2020). Specifically, Wang et al. (2021) used a U-Net model to identify the invasive plant *Solanum rostratum*, achieving a precision of 89.95 % and a recall rate of 90.3 %. A calibration plate was used to calculate the actual ground area corresponding to each UAV image pixel, enabling accurate estimation of *Solanum rostratum* coverage. James and Bradshaw (2020) used an improved semantic segmentation model to successfully identify the invasive plant *Hakea* from UAV images, achieving an F1-score of 83 % and an accuracy of 96 %. Amarasingam et al. (2024) employed the U-Net semantic segmentation model to map the distribution of the invasive Broad-leaved Pepper along coastal regions. Comparative analysis demonstrates that the U-Net model significantly outperforms traditional machine learning models, achieving a precision of 86 %, recall of 76 %, and an F1-score of 81 %, underscoring its efficacy in environmental monitoring applications. Hao et al. (2024) attempted to monitor the growth dynamics of the aquatic invasive plant *Pistia stratiotes* L. using drones. They employed the semantic segmentation network DeepLabV3+ model with ResNet-34, achieving excellent performance. The model yielded an average accuracy of 90.24 % for water lettuce detection across five different dates, with F1-scores ranging from 85.33 % to 96.54 %. These studies demonstrate the reliable performance of semantic segmentation networks in achieving pixel-level segmentation of invasive plants. They also highlight the potential for precise monitoring of plant growth and conducting ecological indicator assessments.

After obtaining high-resolution images captured by drones, these images need to be processed to generate a Digital Orthophoto Map (DOM). Subsequently, the deep learning networks mentioned earlier are employed to classify and predict species, generating a species distribution map (Xu et al., 2023). This can provide researchers and farm management professionals with precise geographical location, coverage area, and coverage density information, facilitating further ecological analysis and farmland management (Xu et al., 2023). However, the current applications of species distribution maps remain limited (Konowalik and Nosol, 2021). Current research largely focuses on calculating coverage, estimating crop yields, and counting fruits and inflorescences (Garcia Millan et al., 2020; Lin et al., 2021; Zhang et al.,

2023). Few studies have applied these maps to species distribution models, niche models, intraspecific relationship analysis, and interspecific competition intensity analysis (Norberto et al., 2023; Reckling et al., 2021; Zhang et al., 2022). The difficulty in directly obtaining substantial ecological and agronomic information from these maps stems from the challenges of integrating high-resolution spatial data with complex spatial models. This study addresses these challenges by utilizing high-resolution species distribution maps to investigate the distribution patterns and spatial autocorrelation of invasive species, thereby providing a basis for precise pesticide application.

Thus, this study aims to construct a stable and reliable semantic segmentation model that effectively identifies these two invasive plants. The model's performance is evaluated across test sets at different altitudes, various categories, and multiple evaluation metrics. Using this model, a high-resolution species distribution map is obtained. The Moran's Index is used to calculate the spatial autocorrelation of the two invasive plants, evaluating their intra-species distribution patterns and finding distribution centers. This lays the groundwork for further precise weeding.

2. Materials and methods

2.1. Study area

The study was conducted in Yantai City, Shandong Province, China, a significant apple-producing region. The exact location of the research area is at 37°12'37.8510"N latitude and 120°35'40.5012"E longitude (Fig. 1). The tree row spacing of the orchard is 3.8 m × 1.5 m, and the trees range from 4 to 5 m in height.

Before the experiment, a preliminary survey of the orchard weed community diversity was conducted. The dominant species identified were *Erigeron annuus* (L.) Pers., *Erigeron canadensis* L., *Commelina communis* L., *Alopecurus aequalis* Sobol., *Cynanchum rostellatum*, *Ipomoea nil*, and *Rorippa indica* (L.) Hiern. Among these, *Erigeron annuus* (L.) Pers. and *Erigeron canadensis* L. are globally recognized invasive species, both

belonging to the Asteraceae family. These species are known to produce allelopathic chemicals, which inhibit the growth of other plants, giving them a competitive advantage. Additionally, they exhibit herbicide resistance (Peruzzi et al., 2023; Rana et al., 2023).

The drone survey was conducted in July 2022, when the fruit trees were in the fruit enlargement stage, characterized by a spindle-shaped canopy structure, while both *E. annuus* and *E. canadensis* were in flowering stage. The orchard features flat terrain and uniform brown earth conditions. In July 2022, the area recorded a total monthly rainfall of 217 mm. The average monthly conditions were as follows: 84.31 % relative humidity, 25.72 °C air temperature, 52.42 % soil moisture, 25.33 °C soil temperature, and 19.77 klux light intensity. The orchard was managed with standard agricultural practices such as regular irrigation, pest control, and nutrient management to ensure optimal growth conditions. Weather conditions during data collection were generally stable, with no extreme weather events reported, ensuring the consistency and reliability of the data collected.

2.2. UAV image acquisition

Data collection was performed using a DJI M300 RTK drone (SZ DJI Technology Co., Ltd., China). RGB images were captured at a resolution of 8192 × 5460 pixels. The drone was equipped with a DJI DL 35 mm F2.8 LS ASPH lens, and the sensor size was 35.9 mm × 24 mm. To retain more image details and minimize background blur, the aperture was set to f/16 during shooting. The flight path was automatically planned to use the mapping and aerial photography features of the DJI Pilot App (SZ DJI Technology Co., Ltd., China). To ensure high-quality Digital Orthophoto Map (DOM), the flight direction overlap was set to 80 % and the lateral overlap to 70 %, with a flight speed of 1 m/s. The Real Time Kinematics (RTK) feature was activated throughout the flight, and the geographic projection coordinates were set to WGS84. Table 1 provides detailed information on the drone flight path. Flights were conducted between 10:00 and 14:00 to minimize the impact of sunlight and shadows. Images were collected twice in July 2022. A total of 786

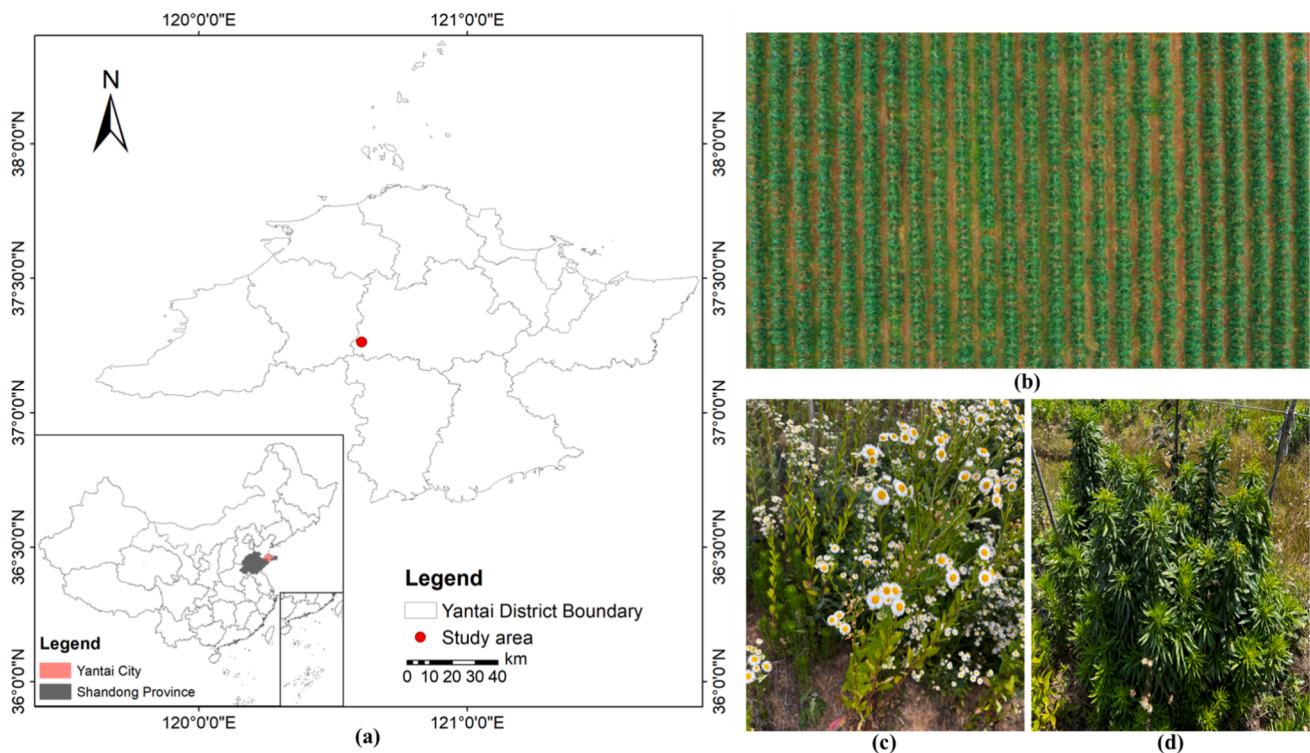


Fig. 1. Location of the study area: (a) the study area in Yantai City, Shandong Province, China; (b) UAV image of the study area; (c) mobile photo of *Erigeron annuus* (L.) Pers.; (d) mobile photo of *Erigeron canadensis* L.

Table 1
UAV flight parameters and corresponding data collection metrics.

Flight Altitude (m)	GSD (cm/pixel)	Flight Path Length (m)	Flight Duration (min)	Number of Images
5	0.0625	651	11	954
10	0.1250	326	6	238
15	0.1875	217	4	106
20	0.2500	163	3	60
25	0.3125	130	2	38

images were captured under clear, cloudless conditions on July 8, with an average light intensity of 84.44 ± 5.36 klux. To enhance the model's robustness, additional data were collected from the same plot under cloudy conditions on July 31, resulting in 214 images with an average light intensity of 51.90 ± 13.67 klux.

The flight altitudes were 5 m, 10 m, 15 m, 20 m and 25 m, corresponding to Ground Sampling Distances (GSD) of 0.0625 cm/pixel, 0.125 cm/pixel, 0.1875 cm/pixel, 0.25 cm/pixel, and 0.3125 cm/pixel, respectively. The method for calculating GSD is outlined in Eq. (1).

$$GSD = (H \times S_w) / (F \times Im_w) \quad (1)$$

where GSD is ground sampling distance in centimeters/pixels, H is height in meters, S_w is sensor width in millimeters, F is focal length in millimeters, Im_w is image width in pixels.

The acquired images were processed using the software ContextCapture v.10.20 (Bentley Systems, United States), to construct the Digital Orthophoto Map (DOM) and Digital Surface Model (DSM) of the experimental area. This process included importing ground control points (GCPs), building an aerotriangulation model, filtering blurry images, matching images, and creating a dense point cloud. The geographic projection coordinate system used was WGS84 UTM Zone 51 N.

2.3. Data preparation and generation

The preparation of the training, test, and validation sets involved the following steps: Firstly, the original images (8192×5460 pixels) were randomly cropped into smaller images of 512×512 pixels. This method facilitated effective neural network training and reduced computational resource consumption, without significantly losing detail by resizing the original images. At each of the five altitudes (5 m, 10 m, 15 m, 20 m, and 25 m), 200 images of 512×512 pixels were obtained. Subsequently, the Labelme tool, an open-source image annotation software, was used to

annotate these images, with categories for *E. annuus*, *E. canadensis*, and apple tree. The annotations were done to include the edges of the plants as much as possible.

To enhance the model's generalization capabilities, nine types of data augmentation were performed on this dataset: flip, rotation, elastic deformation, contrast adjustment, brightness adjustment, sharpening, noise addition, blurring, and mosaic addition. Fig. 2 shows examples for each augmentation type. This process also simulated variations in ISO, aperture, shutter speed, shooting angle, target occlusion, and lighting conditions. These augmentations enabled the trained model to better generalize and improve recognition in complex field scenarios (Schiefer et al., 2020).

After augmentation, the total dataset consisted of 10,000 images. It was divided into training, validation, and test sets in a 70:20:10 ratio. This resulted in 7,000 images for training, 2,000 images for validation, and 1,000 images for testing. For each individual altitude, there were 1,400 images for training, 400 images for validation, and 200 images for testing. The weed dataset was annotated in Visual Object Classes (VOC) dataset format. This format is widely used and well-supported in semantic segmentation tasks. It facilitates the integration of our dataset with existing tools and models. Additionally, it allows for clear and organized annotation of plant boundaries, making it suitable for our weed identification task.

2.4. Semantic segmentation model

2.4.1. DeepLabV3+ network structure

The network structure used in this study is illustrated in Fig. 3, and the model is based on DeepLabV3+. The model consists of an encoder and a decoder, which can fuse information from multiple scales to classify images at the pixel level, enhancing the accuracy of leaf blade edge recognition and overall feature extraction (Chen et al., 2017).

The encoder extracts the key features and abstract information from the image, incorporating Atrous convolution to precisely control feature resolution. This technique expands the model's receptive field without losing information, enabling it to capture multi-scale details. Xception was chosen as the backbone of the network to increase the model's depth and receptive field (Chollet, 2017).

Xception consists of a linear stack of depthwise separable convolutions connected by residuals, reducing the model's parameters and computation. However, it retains strong feature extraction capabilities. As a backbone, Xception introduces residual connections similar to ResNet, which help mitigate the vanishing gradient problem in deep network training and promote effective information transfer between

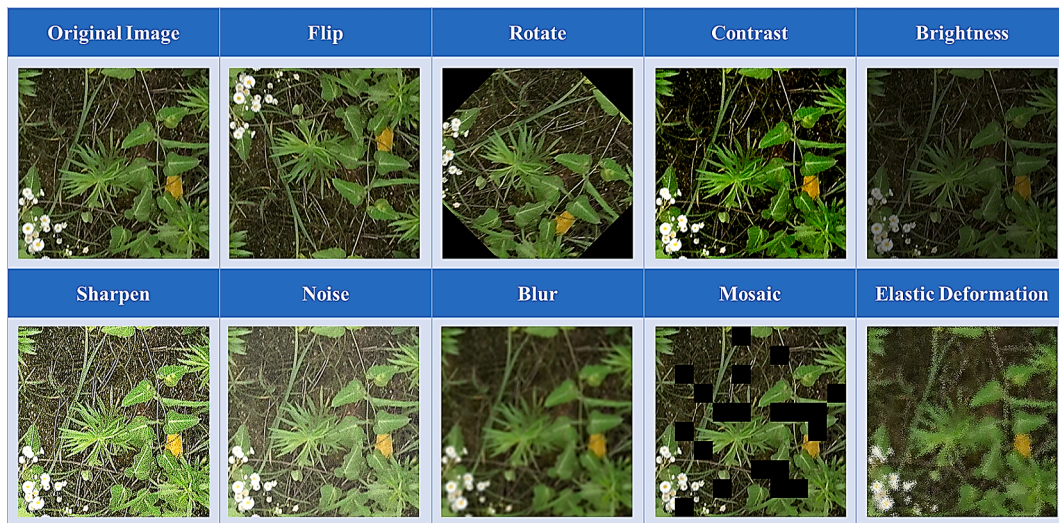


Fig. 2. Effects of various data augmentation methods on image samples.

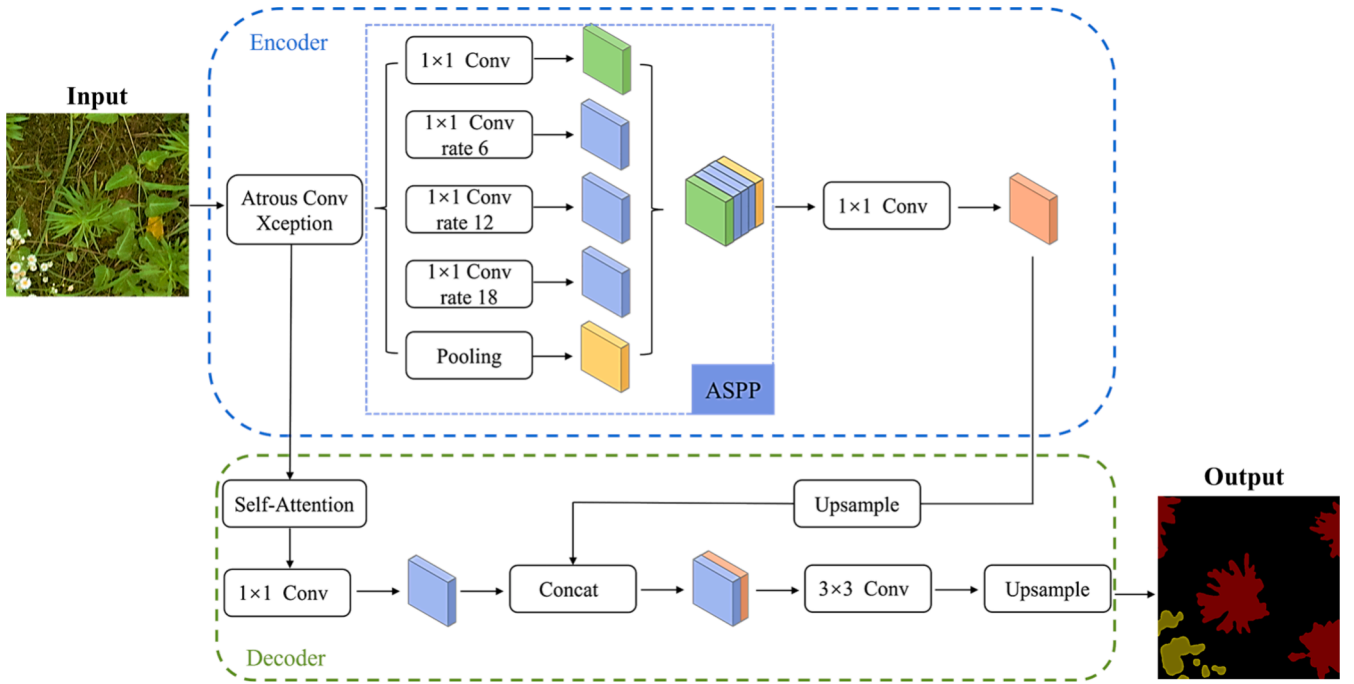


Fig. 3. Architecture of DeepLabV3+ with Xception backbone enhanced by Self-Attention.

layers. While MobileNet is more efficient in terms of computational resources and is suitable for mobile and embedded vision applications, Xception's deeper architecture and more complex residual connections provide better feature extraction and higher accuracy. Xception is divided into three flows: Entry flow, Middle flow, and Exit flow, consisting of 4, 8, and 2 blocks, respectively. This three-stage structure, along with depthwise separable convolutions, aids in extracting rich features while maintaining computational efficiency. Xception strikes a balance between depth, efficiency, and computational cost, making it a suitable choice as the encoder in our model. It can extract a diverse range of features with fewer parameters, aligning well with the goals of this study and providing an optimal foundation for building a high-performance network. Two valid feature layers are produced after Xception processing, one of which is a low-level feature sent straight to the decoder. The Atrous Spatial Pyramid Pooling (ASPP) network is provided with the additional valid feature layer.

The Atrous Spatial Pyramid Pooling module consists of parallel Atrous Convolution, with different rates, constructing convolution kernels with different sensory fields, to obtain multi-scale and richer semantic information. The generated 5 feature layers are merged. Finally, 1×1 convolution is applied to compress the features, resulting in a new feature layer with 256 channels.

The decoder part is used to combine the semantic information of the low-level feature and the high-level feature, which further improves the accuracy of the segmentation boundary information. The high-level features generated by the ASPP module, after up-sampling using bilinear interpolation, matches the output size of the low-level features from the Xception network. Subsequently, the low-level and high-level features are fused to generate a comprehensive feature map. Finally, the fused feature maps are subjected to 3×3 convolution and up-sampling operations, to further reduce the spatial details to obtain the prediction maps.

2.4.2. Self-Attention mechanism

The network architecture employed in this study includes a Self-Attention mechanism module integrated within the encoder (Fig. 4). This module is specifically designed to enhance the processing of low-level feature representations. Its primary function is to capture

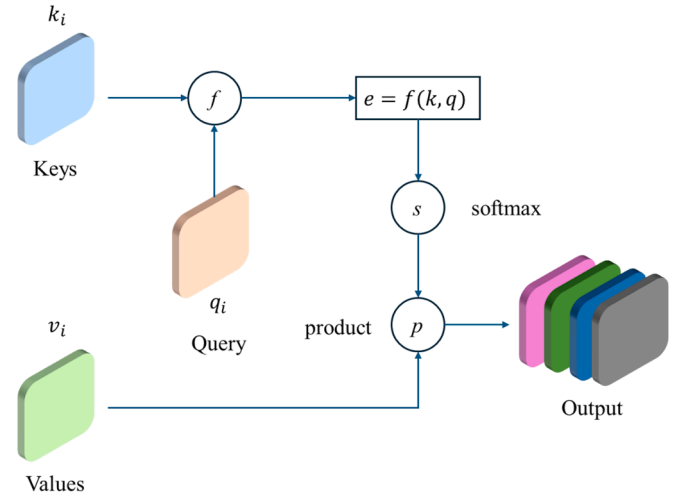


Fig. 4. Architecture of the Self-Attention module.

interrelationships and dependencies among different local regions within the image. By leveraging these relationships comprehensively during training, the model achieves precise pixel-level recognition (Zhao et al., 2020). The implementation of the Self-Attention module enhances the model's capacity to capture detailed information of target plants, including leaf edges, colors, pixels, and gradients. This capability is crucial for addressing the complexities encountered in field conditions where shooting conditions may vary.

The Self-Attention module assigns a feature vector to each pixel or region, indicating its position within the feature space. This model calculates weights for every pair of elements in the sequence, determining the relevance or significance between them. For each local area or pixel in the image, specific Query (Q), Key (K), and Value (V) vectors are generated. These vectors are derived by multiplying the area's or pixel's feature vector with distinct weight matrices.

$$Q_i = X_i W_Q \quad (2)$$

$$K_i = X_i W_K \quad (3)$$

$$V_i = X_i W_V \quad (4)$$

where X_i is the feature vector of region i in the image, and W_Q , W_K , and W_V are the learnable weight matrices corresponding to Q , K , and V , respectively.

For each pixel or region i , we also need to compute the attention scores with all other pixels or regions j in the image. The attention score is calculated by the dot product of Q and K . The formula is:

$$\text{Score}(i, j) = Q_i K_j^T / \sqrt{d_k} \quad (5)$$

Subsequently, we need to pass through the *softmax* function to obtain the probability distribution of the weights, $\text{Weight}(i, j)$, ensuring the sum of the weights is 1. The specific formula is as follows:

$$\text{Weight}(i, j) = \text{softmax}(\text{Score}(i, j)) \quad (6)$$

Finally, the result, Output_i , is obtained by multiplying $\text{Weight}(i, j)$ (the probability distribution of the weights) with V_j (the weight) and summing up. The specific formula is as follows:

$$\text{Output}_i = \sum_j \text{Weight}(i, j) \times V_j \quad (7)$$

In this way, the model can capture the contextual information within the image, obtaining precise expressions through learning, whether the dependencies are local or global. This method is especially crucial in weed identification because it helps the model understand the semantic information of the plants more accurately in complex visual environments (Chen et al., 2022).

2.4.3. Model evaluation metrics

To address the phenomenon of class imbalance when performing multi-category classification of plants, a combination of Dice Loss and Cross-Entropy Loss is employed as the loss function. This approach is chosen for its ability to enhance semantic segmentation performance, particularly in scenarios with uneven class distribution or where precise pixel-level classification is crucial (Zhang et al., 2021). This hybrid approach combines global class distribution with pixel-level overlap, providing a more comprehensive solution for the task's complexity. Unlike using Cross-Entropy Loss (CE Loss) alone, this combined method offers increased robustness in managing class imbalance by leveraging Dice Loss. Dice Loss is derived from the dice coefficient, which is a metric function used to evaluate the similarity between two samples, with values ranging from 0 to 1. A larger value indicates greater similarity. The definition of the dice coefficient is as follows:

$$\text{Dice} = 2|X \cap Y| / (|X| + |Y|) \quad (8)$$

where $|X|$ and $|Y|$ represent the number of elements in matrices X and Y , respectively. The formula for Dice Loss is as follows:

$$\text{DiceLoss} = 1 - \text{Dice} = 1 - 2|X \cap Y| / (|X| + |Y|) \quad (9)$$

This study uses three commonly used performance evaluation metrics: mean intersection over union (mIoU), mean Pixel Accuracy (mPA), and mean Class Precision (mPrecision), as expressed in Eqs. (10–12).

$$\text{mIoU} = (1/C) \times \sum_{i=0}^C TP_i / (TP_i + FP_i + FN_i) \quad (10)$$

$$\text{mPA} = (1/C) \times \sum_{i=0}^C TP_i / (TP_i + FN_i) \quad (11)$$

$$\text{mPrecision} = (1/C) \times \sum_{i=0}^C TP_i / (TP_i + FP_i) \quad (12)$$

where C represents the number of classes, TP (True Positives) refers to the pixels correctly classified as the class of interest, FP (False Positives) represents the pixels incorrectly classified as the class of interest, TN

(True Negatives) denotes the pixels correctly classified as background, and FN (False Negatives) indicates the pixels that were incorrectly classified as background or another class.

These evaluation metrics are commonly employed together to assess the performance of semantic segmentation tasks comprehensively (Guo et al., 2018). Among them, mean Intersection over Union (mIoU) holds particular importance as it evaluates both pixel-level accuracy and inter-class relationships simultaneously. Meanwhile, mean Precision (mPrecision) provides insights into the performance of individual categories, and mean Pixel Accuracy (mPA) measures overall pixel-level accuracy. These metrics facilitate a thorough evaluation of model performance by comparing predicted images with ground truth.

2.4.4. Implementation details

This model underwent training for a total of 200 epochs. During the initial phase of the first 50 epochs, the pre-trained layers were frozen, meaning their weights were kept constant, while only the newly added layers were trained. This strategy preserves the beneficial features from the pre-trained model and helps the model adapt to the new task, enhancing training speed and preventing overfitting. In the subsequent 150 epochs, all layers were unfrozen, allowing the entire network, including the pre-trained layers, to be fine-tuned. This adjustment enables a more precise refinement of the model to better suit the specific dataset, potentially improving performance. The initial learning rate was set to 0.0005, and the minimum learning rate was set to 1 % of the initial rate. A momentum of 0.9 was used, and the learning rate decay followed a cosine annealing schedule. The downsample factor was set to 16. The training process utilized mixed precision (fp16) to speed up training and reduce memory usage. The batch size was set to 16 during the freeze phase and 8 during the unfreeze phase. Training was conducted on an NVIDIA GPU (GeForce RTX 4090 Ti, 24 GB RAM), with code implemented using PyTorch (1.7.1) and Python (3.7.4). The FPS during training was 24.36 iterations per second. The model comprised 55.01 M parameters, with a total computation of 88.27 GFLOPS, balancing model complexity and computational efficiency. Results were recorded every 5 epochs to monitor progress and adjust training as needed.

The hyperparameters for this study were determined through a combination of manual tuning, empirical adjustments, and insights from previous research. The initial learning rate and minimum learning rate were selected based on preliminary experiments. The batch sizes were chosen considering the memory constraints of our training hardware and the need to balance training speed and model performance.

2.5. Vegetation coverage and Moran's Index

The Digital Orthophoto Map (DOM) images were cropped uniformly to 512×512 pixels, maintaining their geographical integrity. These cropped images were processed through the model to generate prediction images of the target plants. Subsequently, using the geographic information associated with each image, the predicted images were integrated back together to create the plant distribution map (Fig. 5).

Based on the number of pixels of different species, the coverage rate and coverage area of the target plants were calculated. The specific formula is as follows:

$$\text{CoverageRate} = N_i / N_{\max} \times 100\% \quad (13)$$

where N_i represents the number of pixels of category i in the image of the research area, and $N_{\max}$ is the total number of pixels in the image of the study area.

$$S_i = N_i \times GSD^2 \times 10^{-4} \quad (14)$$

where GSD is ground sampling distance in centimeters/pixels, and S_i represents the coverage area of plant i in square meters.

The plant distribution map was analyzed in Moran's Index (Moran's

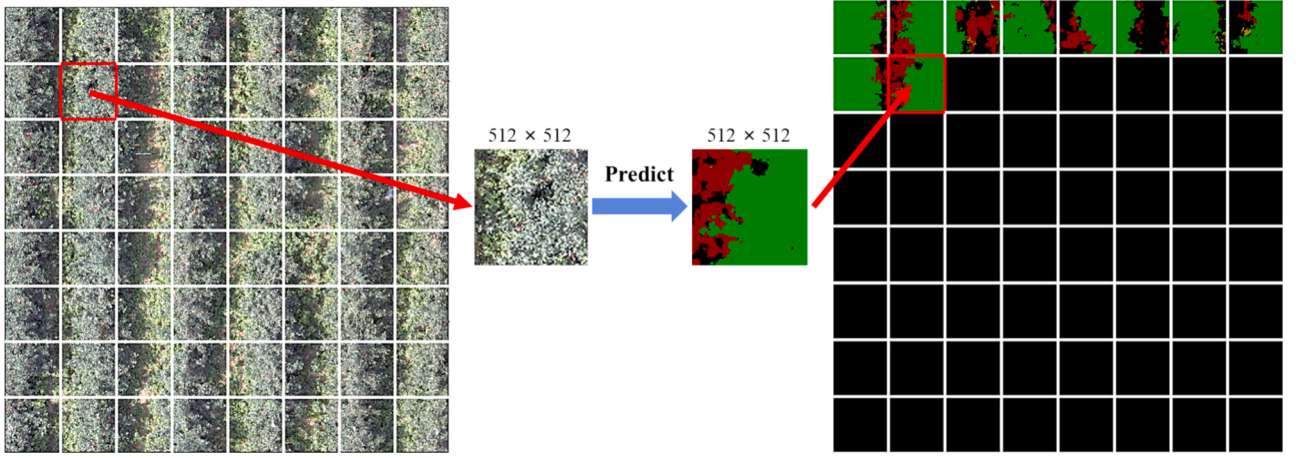


Fig. 5. Schematic diagram of the invasive plant distribution map production process.

I), a statistical method employed to study spatial distribution patterns within geographic datasets. This analysis was conducted through ArcGIS Pro 3.0.0 (ESRI, Redlands, CA, USA). Moran's Index is particularly useful for examining how plants are spatially distributed across an area of interest (Lee and Li, 2017). Moran's Index aids in assessing if there is spatial autocorrelation among plant communities and quantifies the extent of this correlation. This analysis determines whether the distribution of target plants shows clustering or dispersion across the study area. The global Moran's Index provides a measure of spatial autocorrelation for the entire geographical dataset, offering insights into the spatial pattern exhibited by the target plants. The formula for its calculation is as follows:

$$I = (N/W) \times \sum_{i=1}^N \sum_{j=1}^N w_{ij} (x_i - \bar{x})(x_j - \bar{x}) / \sum_{i=1}^N (x_i - \bar{x})^2 \quad (15)$$

where I is the global Moran's Index, N is the total number of geographical units in the study area, and W is the total number of elements in the adjacency matrix, representing the connections between neighboring geographical units. w_{ij} is the connection weight of geographical units i and j in the adjacency matrix. x_i and x_j are the attribute values of the target plant, and $\bar{x}$ is the mean value of the attribute values in the research area.

Anselin's Local Moran's I is utilized to examine local spatial autocorrelation at each location (Yamada, 2024). This can generate a Local Moran Scatterplot, which illustrates the type of spatial correlation for each geographic unit: "High-High (HH)", "Low-Low (LL)", "High-Low (HL)", and "Low-High (LH)". In "HH" scenarios, a region and its neighboring areas all exhibit high values, indicating a cluster of high values or a hotspot area. Conversely, "LL" denotes that a region and its neighbors have uniformly low values, depicting a cluster of low values or a cold spot area. "HL" signifies a high-value region surrounded by low-value areas, identifying it as a spatial outlier with high-value anomaly. "LH" indicates that a low-value region is surrounded by high-value areas, representing a low-value anomaly. Regions labeled as "No significant" lack significant spatial autocorrelation with their surroundings, indicating insufficient evidence to support notable similarity or connection in spatial terms.

$$\text{Anselin's Local Moran's } I_i = ((x_i - \bar{x})/S_i^2) \times \sum_{j=1, j \neq i}^N (w_{ij}(x_j - \bar{x})) \quad (16)$$

where S_i^2 is the variance of the plant attribute values of all geographic units, w_{ij} is the connection weight of geographical units i and j in the adjacency matrix. x_i and x_j are the attribute values of the target plant, and $\bar{x}$ is the mean value of the attribute values in the research area.

When performing Moran's index operations, the DOM image must first be resampled. Following the standards of field sampling surveys,

the smallest geographical unit was set at 0.1 m, which was also close to the individual covering area of *E. annuus* and *E. canadensis* plants. The resampling method was suitable for the Nearest Neighbor Resampling method of the vegetation analysis. Subsequently, the raster calculator extracted pixel points corresponding to the target plants, converting the resultant raster data into point data. When calculating Moran's I , the distance method was Euclidean. Moran's I was calculated using the Euclidean distance method, with a distance threshold of 1 m, based on field survey standards. Spatial relationships were modeled using inverse distance. For Anselin Local Moran's I , statistical significance was determined using a false discovery rate (FDR) correction at a 95 % confidence level. The analysis involved 499 permutations to ensure robust statistical assessment.

3. Results

3.1. Model ablation test performance

This study conducted ablation experiments on three models: (1) DeepLabV3+ with MobilenetV2 as the baseline; (2) the baseline model with Xception as the backbone; and (3) the proposed model, which integrates Self-Attention into the decoder of DeepLabV3+ Xception (Table 2). The results of these models' mIoU reflected their performance differences in detecting various categories and heights.

In the baseline model (DeepLabV3+ with MobileNetV2), high segmentation accuracy was observed for the Apple Tree class across distances, with a peak mIoU of 98.52 % at 5 m, maintaining 96.53 % at 25 m. However, performance on *E. canadensis* and *E. annuus* was lower, particularly for *E. annuus*, which dropped from 88.67 % at 5 m to 73.26 % at 25 m, indicating distance-related challenges in segmenting certain vegetation classes.

With an Xception backbone, the DeepLabV3+ model improved segmentation accuracy for *E. canadensis* and Apple Tree classes, achieving a consistent mIoU for *E. canadensis* from 97.27 % at 5 m to 92.50 % at 25 m, and maintaining 97.44 % for Apple Tree at 25 m. *E. annuus* also showed better performance than with MobileNetV2, though it still declined from 90.68 % at 5 m to 77.96 % at 25 m.

The proposed model, which integrates self-attention with the Xception backbone, outperformed all configurations across classes and distances. It achieved an mIoU of 99.03 % at 5 m for Apple Tree, sustaining 98.19 % at 25 m, and improved resilience for *E. canadensis* with an mIoU of 95.03 % at 25 m. *E. annuus* saw the largest improvement, maintaining 85.43 % at 25 m. These results highlight the proposed model's robustness and suitability for plant species segmentation across varying distances.

Qualitative results are presented in Fig. 6, displaying the model's

Table 2
Comparisons of ablation study.

Model	Class	mIoU (%)				
		5 m	10 m	15 m	20 m	25 m
DeepLabV3+_MobilenetV2	<i>E. canadensis</i>	96.38	92.80	93.28	88.52	89.53
	<i>E. annuus</i>	88.67	80.66	76.87	68.27	73.26
	Apple Tree	98.52	97.97	96.88	97.63	96.53
DeepLabV3+_Xception	<i>E. canadensis</i>	97.27	94.56	95.24	92.53	92.50
	<i>E. annuus</i>	90.68	84.64	81.93	85.02	77.96
	Apple Tree	98.85	98.51	97.75	98.46	97.44
DeepLabV3+_Xception_Self-Attention (proposed)	<i>E. canadensis</i>	98.02	96.16	96.74	95.10	95.03
	<i>E. annuus</i>	92.68	87.44	85.65	88.93	85.43
	Apple Tree	99.03	98.88	98.25	98.84	98.19

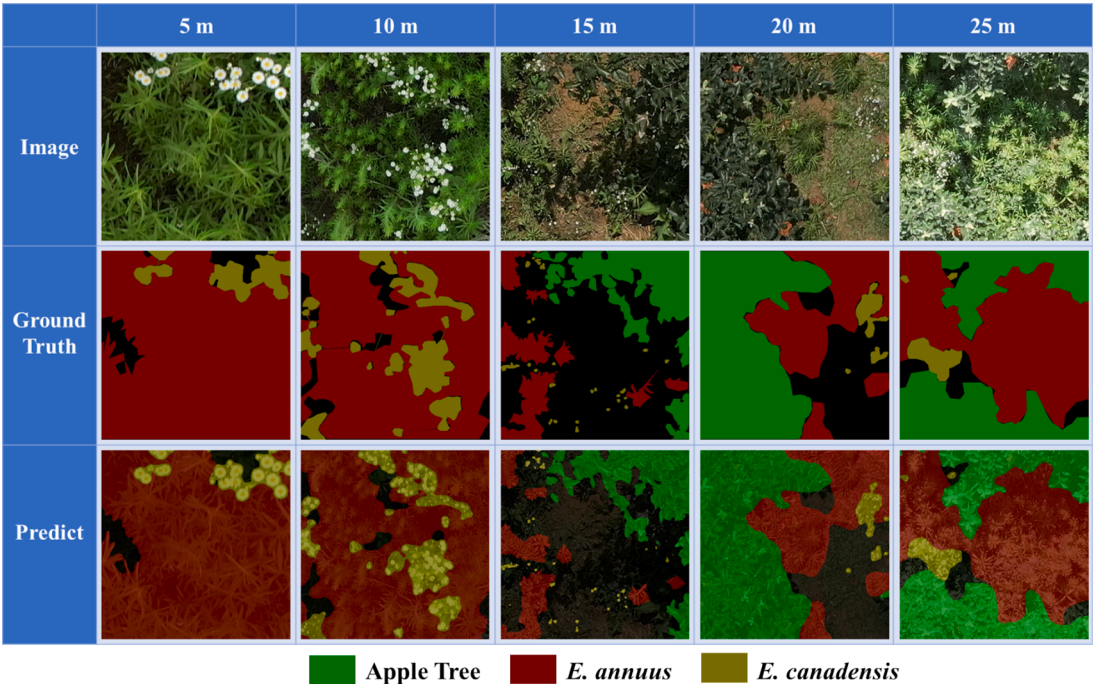


Fig. 6. Segmentation results of the proposed model at different heights.

predictions alongside ground truth annotations. At a height of 5 m, the model accurately segments *E. canadensis* and *E. annuus*, closely matching the ground truth, which indicates strong performance at close range. At 10 m, the model maintains high accuracy with only minor deviations, particularly in distinguishing *E. annuus* features. By 15 m, more noticeable differences appear, with some blending observed between *E. canadensis* and *E. annuus*. At 20 m, the model exhibits increased misclassification for smaller plant classes, showing a decline in accuracy with distance. At the furthest distance (25 m), Apple Tree maintains segmentation accuracy, but smaller classes show more blending, reflecting limitations in resolution due to distance.

3.2. Comparison with other semantic segmentation models

The mIoU performance of this model compared to other models is shown in Fig. 7. The minimum value is 80.39 %, which is the result of SegNet at 25 m, and the maximum value is 96.05 %, which is the result of proposed model at 5 m. As the height decreases, the mIoU of all four models exhibits different increases. At the same height, the mIoU of this model is consistently higher than the other three models.

As shown in Fig. 8, the performance of the proposed model is compared with three benchmark semantic segmentation models (SegNet, FCN, and UNet) across key metrics: mIoU, mPrecision, and mPA. The test set includes a mix of all classes and images at various heights,

providing a comprehensive assessment of each model’s generalization capabilities. The proposed model demonstrates superior performance across all metrics, achieving the highest scores in mIoU, mPrecision, and mPA. Specifically, it reaches an mIoU above 94 %, surpassing SegNet, FCN, and UNet, which score approximately 88 %, 92 %, and 92 %, respectively. The proposed model also excels in mPrecision and mPA, with values consistently exceeding 95 %, indicating more precise and accurate segmentation compared to the benchmark models. Overall, these results highlights the robustness and accuracy of the proposed model, underscoring its effectiveness in handling a variety of classes and distances.

3.3. Vegetation coverage

Fig. 9 displays the DOM image of the test area and the Invasive Plant Distribution Map predicted by the model. The weed map shows the invasive plants’ specific locations and distribution range.

The pixel calculation obtained are presented in Table 3. The coverage rates for the two invasive plants, *E. canadensis* and *E. annuus*, are 16.13 % and 0.49 %, respectively. The specific coverage areas are 70.02 m² and 2.14 m², respectively.

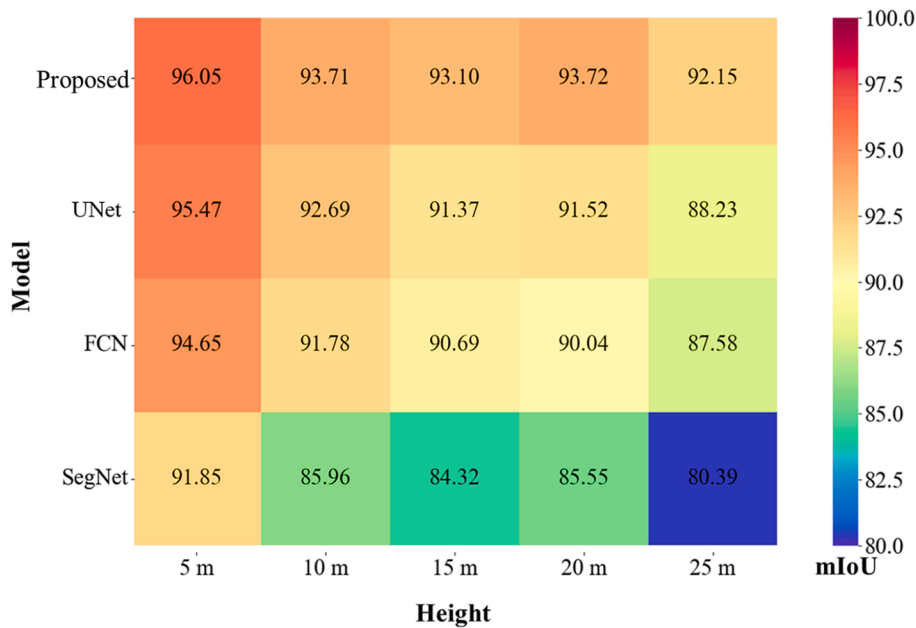


Fig. 7. Miou performance comparison of various models on test sets at different heights.

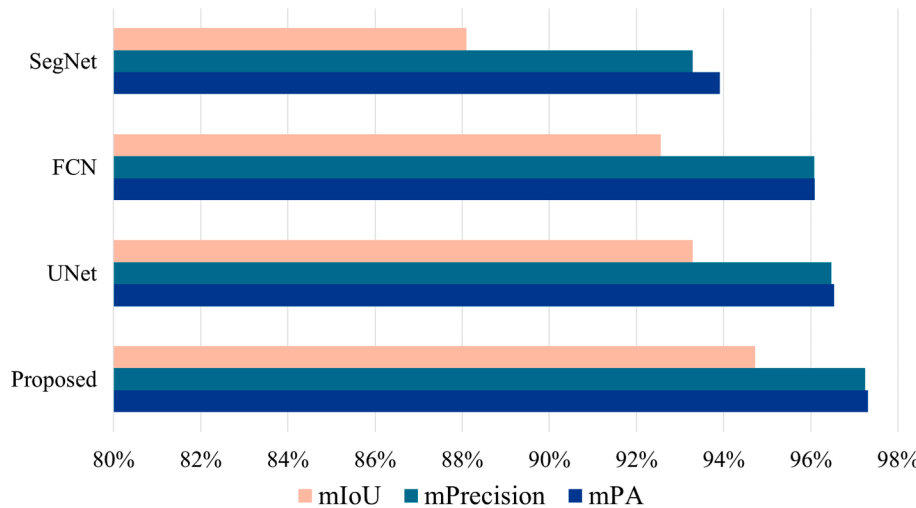


Fig. 8. Comparison of segmentation model performance on mixed-height test set.

3.4. Moran's Index

The Global Moran's Index for *E. annuus* is 0.03, with a Z-score of 84.82 and a p-value less than 0.05. The Global Moran's Index for *E. canadensis* is 0.23, with a Z-score of 547.67 and a p-value less than 0.05. The likelihood of random distribution for the two plants is less than 1 %, both exhibiting a clustered distribution.

Fig. 10 presents the results of spatial clustering analysis using Anselin Local Moran's I for *E. canadensis* and *E. annuus*, highlighting patterns of high and low density distribution that reveal distinct clustering characteristics of these two species. The clustering map for *E. canadensis* (Fig. 10a) shows significant spatial autocorrelation, with prominent high-high and low-low clusters. High-high clusters (in red) mark areas with notably high densities, suggesting potential hotspots of dense growth. Low-low clusters (in dark blue) indicate regions with consistently low densities. High-low (pink) and low-high (light blue) clusters are relatively rare, suggesting limited transitional zones between high and low density areas. This spatial pattern implies that *E. canadensis* grows in dense, well-defined patches, with clear spatial separation from

low-density areas. In contrast, the clustering map for *E. annuus* (Fig. 10b) reveals a more dispersed distribution, with fewer high-high clusters. The prevalence of low-low clusters (dark blue) indicates an overall lower density, with only sparse areas of higher density. While there are isolated high-high clusters, representing small pockets of dense growth, these are infrequent. This distribution pattern suggests that *E. annuus* does not form large, continuous patches like *E. canadensis*, but rather exhibits a more scattered growth pattern with limited high-density areas.

These results underscore distinct spatial growth behaviors between two invasive plants. *E. canadensis* displays dense, well-defined clusters, suggesting a tendency for localized, high-density growth. In contrast, *E. annuus* is more sparsely distributed, with fewer high-density clusters, indicating a more scattered spatial pattern. This analysis provides insights into the ecological behavior and habitat preferences of each species, which could inform targeted management strategies based on their clustering tendencies.

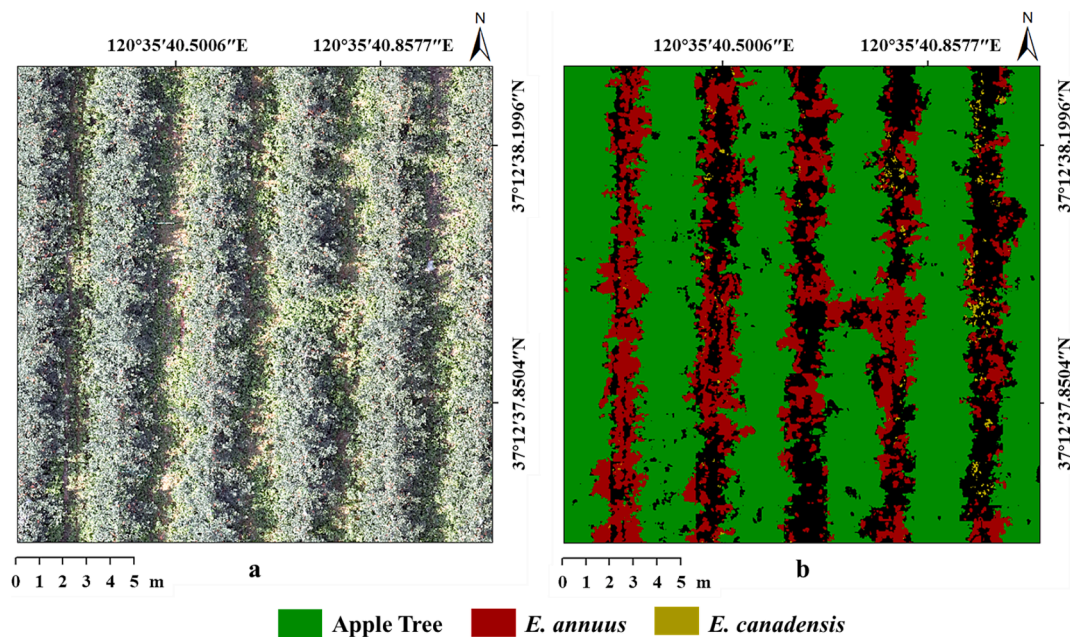


Fig. 9. (a) DOM of the test area and (b) predicted invasive plant distribution map.

Table 3
Coverage area and coverage rate of invasive plants.

Class	Number of pixels	Coverage area (m ²)	Coverage rate (%)
<i>E. canadensis</i>	19,917,798	70.02	16.13
<i>E. annuus</i>	607,416	2.14	0.49
Apple Tree	76,282,442	268.18	61.78

4. Discussion

High-resolution distribution maps of invasive plants can provide farm managers and with researchers precise spatial distribution information, which is crucial for precision weed control. Previous studies indicate that high-resolution remote sensing images offer models more

detailed features, yielding clearer plant boundaries in the maps (Almalki et al., 2022; James and Bradshaw, 2020; Martinez et al., 2021). However, the capacity of models to process detailed information and integrate multi-scale features still requires improvement (Li et al., 2016; Yu et al., 2019a; Yu et al., 2019b). Therefore, we developed a new model to test its detection capabilities on images with resolutions ranging from 0.0625 to 0.3125 cm/pixel. The integration of self-attention mechanisms within the decoder of the DeepLabV3+ model with xception backbone significantly enhanced segmentation performance across all evaluated classes and distances. The Self-Attention module effectively bolstered the model’s capability to focus on relevant spatial details, allows the model to better capture and retain critical features across distances, which improved feature distinction and segmentation consistency even at extended distances. The improved resilience to

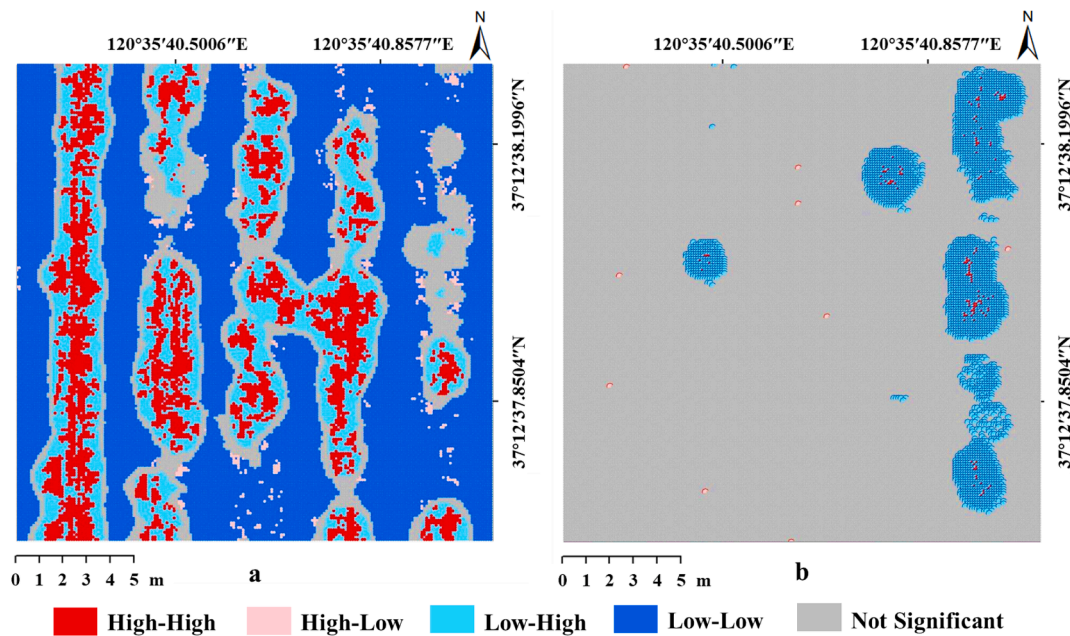


Fig. 10. Anselin Local Moran's I clustering map. (a) *Erigeron canadensis* L. clustering map; (b) *Erigeron annuus* (L.) Pers. clustering map.

distance-based performance degradation and ability to generalize across various plant species make the proposed model highly suitable for practical applications, such as automated agricultural monitoring and ecological studies. While the proposed model excels at close distances, some blending and misclassifications emerge as the distance increases, particularly among smaller classes. This observation suggests that while Self-Attention improves resilience to distance, challenges remain in distinguishing closely located classes as the spatial resolution decreases. However, the Apple Tree class retained high segmentation accuracy, suggesting that the model can effectively handle larger and structurally simpler classes despite distance. Future work could explore further optimizations, such as fine-tuning self-attention configurations or investigating other attention mechanisms to refine performance on smaller or more complex vegetation classes further.

Previous research has shown that dataset quality significantly affects model generalization (Chen and Fan, 2021; He et al., 2023). To enhance dataset diversity, this study employed nine distinct data augmentation techniques along with random cropping to simulate plant imagery under varying environmental conditions. Data collection covered multiple plant growth stages, including the bud initiation, full bud, early flowering, peak flowering, end of flowering, and vegetative stages. While this approach successfully increased dataset variability, it also added complexity to manual annotation. Semantic segmentation annotation requires precise delineation of plant edges, a challenging task due to the diverse growth patterns and stages of plants in natural settings. Consequently, the annotation process is notably time-intensive. Future research will investigate automated labeling methods to alleviate annotation workload. In addition, variability in cloud cover significantly impacts the quality of drone-captured images. This study included images taken under cloudy conditions to further enhance dataset diversity. However, future experiments are needed to quantify cloud cover effects, enabling an analysis of the relationships between cloud cover, lighting, and model performance.

Previous studies analyzing species distribution maps have primarily focused on calculating coverage, estimating crop yields, and counting fruits (Luo et al., 2023; Zhang et al., 2023). It is essential to go beyond these metrics and understand plant distribution patterns to target removal efforts effectively. However, analyzing species distribution patterns from high-resolution maps presents substantial challenges, as models must handle vast data volumes and noise while integrating complex image and spatial data (Isaac et al., 2020; Valavi et al., 2022). Using vegetation coverage, Global Moran's Index, and Anselin Local Moran's I, this study reveals important distinctions in the distribution and clustering behavior of the invasive species *E. canadensis* and *E. annuus*, each demonstrating unique ecological characteristics that could influence management strategies. The dense clustering pattern of *E. canadensis* suggests that management efforts could focus on controlling specific high-density hotspots to curb its spread. In contrast, *E. annuus*, with its scattered growth, may require a broader, landscape-level approach to detect and manage its dispersed populations effectively. Tailoring strategies to each species' spatial distribution could enhance control efficacy and resource allocation in managing invasive species impacts on native ecosystems.

When calculating the Moran's index, the size of the resolution after resampling is crucial, as it will affect the assessment of clustering distribution. Based on preliminary field surveys, the minimum bounding rectangle size of individual plants of the two species in the horizontal space is approximately 0.1 m × 0.1 m. Therefore, 0.1 m is chosen as the minimum research distance for the ecological niche and serves as the resolution after resampling. When identifying other plants, this resolution should be adjusted based on the plant's distribution pattern in the horizontal space. Subsequently, SLAM and other technologies can also be employed to enable real-time monitoring of weeds and autonomous precision weeding by drones. Additionally, environmental parameters can also be introduced to construct interspecific competition models.

5. Conclusion

In this study, high-resolution images of the invasive species *Erigeron annuus* (L.) Pers. and *Erigeron canadensis* L. were collected at various heights through drones. A dataset of 10,000 images was built using nine different image augmentation methods. By integrating the Self-Attention Mechanism with DeepLabV3+, a network structure more suitable for the pixel level semantic segmentation of invasive plants was designed. This model was employed to generate high-resolution invasive plant distribution maps, followed by an analysis of their species distribution patterns. The findings are as follows:

- (1) The improved network structure effectively segmented the target invasive plants from actual orchard images taken at different heights, achieving an overall performance of mIoU of 0.947.
- (2) The improved network structure outperformed other models across different performance evaluation metrics, accurately segmenting target plants in complex field environments, demonstrating greater robustness.
- (3) High-resolution species distribution maps were obtained. The coverage rates for *Erigeron canadensis* L. and *Erigeron annuus* (L.) Pers. in the orchard were calculated to be 16.13 % and 0.49 % respectively, which provided a foundation for detailed ecological analysis.
- (4) The intraspecific distribution patterns for both invasive plants were identified as clustering. The Global Moran's Index for *E. annuus* is 0.03, with a Z-score of 84.82, for *E. canadensis* is 0.23, with a Z-score of 547.67. The results were statistically significant.
- (5) Clustering distribution maps for the two invasive plants were generated. The cluster centers were identified, which allows for precise eradication and control measures targeting these centers. This provides a new method for precise weeding and efficient control of invasive plants.

CRediT authorship contribution statement

Jun Chao: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Data curation, Conceptualization. **Kaiwen Wang:** Software, Methodology, Formal analysis, Data curation, Conceptualization. **Beibei Xu:** Software, Methodology, Conceptualization. **Mary Harty:** Methodology, Investigation, Conceptualization. **Wensheng Wang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Kevin McDonnell:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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